

Uncertainty-Aware DC-LSTM for FX Forecasting and Trading

Group 17

Univeristy College London

Abstract. We propose an uncertainty-aware DC-LSTM framework that combines macroeconomic and market features. Predictive uncertainty is estimated with MC Dropout and Gaussian likelihood, and used in an SNR-based trading rule to filter low-confidence signals. Results show that while point prediction accuracy improves only marginally, the incorporation of uncertainty leads to significant gains in trading performance. The proposed model achieves higher return and Sharpe ratio with lower market exposure than deterministic baselines. This shows that uncertainty is more useful for trade selection than for point forecasting in FX markets.

1 Introduction

Forecasting foreign exchange (FX) rates is a fundamental yet challenging problem in financial time series modelling. FX markets exhibit an extremely low signal-to-noise ratio (SNR), where meaningful patterns are often hidden within stochastic fluctuations. As a result, simple baselines such as the random walk model ($\hat{P}_{t+1} = P_t$) are robust and difficult to outperform.

Recent advances in deep learning have improved performance over traditional econometric models for complex time series forecasting. Recurrent architectures such as Long Short-Term Memory (LSTM) models can capture non-linear temporal dependencies, while exogenous inputs can enhance robustness under volatile market conditions [3]. However, challenges such as data scarcity, non-stationarity, and extreme market shocks persist. Prior work has explored transfer learning, signal decomposition, and hybrid model architectures to address these issues [7, 11, 12, 16]. Despite these advances, most approaches fail to provide reliable uncertainty estimates for decision-making [5, 6].

This motivates uncertainty-aware forecasting, as financial decisions depend not only on expected returns but also on associated risks, which cannot be captured by deterministic predictions alone.

To address this, we adopt a probabilistic forecasting framework that outputs predictive distributions rather than point estimates. Our approach combines macroeconomic feature engineering with a regularised deep learning model and employs Gaussian likelihood together with Monte Carlo (MC) Dropout to approximate predictive uncertainty. We evaluate predictive performance, the calibration and robustness of model outputs in a trading context.

2 Methods

2.1 Data & Features

Our dataset comprises macro-level and market-level features [1,2,9,13–15]. Macro variables include global indicators such as the VIX, interest rates, yield series, and commodity prices (e.g., gold and oil), capturing systemic risk. Market variables include GBP exchange rates and equity indices (e.g., the FTSE 100), reflecting short-term dynamics. The dataset contains 2,757 daily observations from February 2015 to December 2025.

To ensure stationarity, price variables are transformed into log returns $\ln(P_t/P_{t-1})$, while yields and rates are expressed as first differences. The prediction target is the **next-day GBP/USD return**. The data is split chronologically (80/20), with preprocessing fitted on the training set to avoid look-ahead bias.

GBP/USD returns exhibit positive skewness and high kurtosis, indicating heavy tails and frequent extremes. Given the low signal-to-noise ratio in FX data, we construct lagged features, rolling means, and rolling volatility to capture short-term dynamics. Feature selection is performed in two stages: removing variables with pairwise correlation above 0.85 and discarding features that do not outperform random noise in importance tests. The final feature set contains 17 variables capturing global risk sentiment, interest rate differentials, and commodity effects.

2.2 Model

To capture interactions between financial indicators, we adopt a Deep Coupled Long Short-Term Memory (DC-LSTM) architecture for exchange rate forecasting [4]. The model separates market-level and macro-level dynamics into two LSTM branches, reflecting the different temporal characteristics of short-term market movements and slower macroeconomic trends.

As shown in Figure 1, the two branches first extract temporal representations from market and macroeconomic features. Their hidden states are then combined and passed to a coupled LSTM layer to model cross-feature interactions. The resulting representation is fed into a fully connected layer to produce the next-step prediction.

For comparison, we also evaluate standard LSTM, Bi-LSTM, and a simplified Dual-Branch LSTM [4] as baseline models.

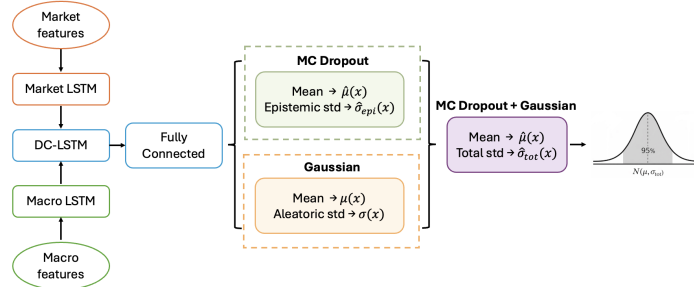


Fig. 1: DC-LSTM Architecture with MC Dropout and Gaussian Likelihood

2.3 Uncertainty Quantification

We quantify predictive uncertainty using MC Dropout for epistemic uncertainty and Gaussian likelihood for aleatoric uncertainty.

MC Dropout: We keep dropout active at test time and run the same input through the network multiple times. Let $f(x; \omega^{(s)})$ be the prediction from the s -th stochastic forward pass. For S samples, the predictive mean and epistemic variance are:

$$\hat{\mu}(x) = \frac{1}{S} \sum_{s=1}^S f(x; \omega^{(s)}), \quad \hat{\sigma}_{\text{epi}}^2(x) = \frac{1}{S} \sum_{s=1}^S \left(f(x; \omega^{(s)}) - \hat{\mu}(x) \right)^2.$$

A 95% prediction interval is $\hat{\mu}(x) \pm 1.96 \hat{\sigma}_{\text{epi}}(x)$. At inference, this is implemented by keeping dropout active.

Gaussian likelihood: To model aleatoric uncertainty, the network predicts both the conditional mean $\mu(x)$ and conditional log-variance $\log \sigma^2(x)$, assuming $y | x \sim \mathcal{N}(\mu(x), \sigma^2(x))$. The conditional density and the negative log-likelihood (dropping the constant term):

$$p(y | x) = \frac{1}{\sqrt{2\pi\sigma^2(x)}} \exp\left(-\frac{(y - \mu(x))^2}{2\sigma^2(x)}\right), \quad \mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\log \sigma^2(x) + \frac{(y - \mu(x))^2}{\sigma^2(x)} \right).$$

We predict $\log \sigma^2(x)$ rather than $\sigma^2(x)$ for positivity and numerical stability, and constrain it by

$$\log \sigma^2(x) = a + (b - a) \text{sigmoid}(z(x)).$$

This smooth bounded mapping prevents unstable interval widths.

Combined epistemic and aleatoric uncertainty: We also evaluate a combined setting by applying MC Dropout to the Gaussian model. Each stochastic pass outputs $\mu^{(s)}(x)$ and $\sigma^{(s)}(x)$. The predictive mean and epistemic variance:

$$\hat{\mu}(x) = \frac{1}{S} \sum_{s=1}^S \mu^{(s)}(x), \quad \hat{\sigma}_{\text{epi}}^2(x) = \text{Var}_s(\mu^{(s)}(x)),$$

and the aleatoric variance and the total predictive variance are

$$\hat{\sigma}_{\text{ale}}^2(x) = \frac{1}{S} \sum_{s=1}^S (\sigma^{(s)}(x))^2, \quad \hat{\sigma}_{\text{tot}}^2(x) = \hat{\sigma}_{\text{epi}}^2(x) + \hat{\sigma}_{\text{ale}}^2(x),$$

which gives the final interval $\hat{\mu}(x) \pm 1.96 \hat{\sigma}_{\text{tot}}(x)$. In the final implementation, the Gaussian model is first set to evaluation mode, with only the explicit dropout layer re-enabled for better calibration.

3 Experiments

3.1 Experimental Setup

We evaluate all models using daily GBP/USD returns covering 550 trading days, following the preprocessing pipeline described in Section 2. Model inputs are constructed using a rolling window of length 30. Predicted log returns are then converted into price levels via $P_{t+1} = P_t \exp(\hat{r}_{t+1})$ (Fig. 2).

To translate predictions into trading decisions, we implement an SNR-based filtering rule to filter low-confidence signals. This mechanism executes trades

only when the predicted signal is sufficiently strong relative to its uncertainty:

$$\text{Signal} = \begin{cases} \text{sign}(\mu) & \text{if } \frac{|\mu|}{\sigma_{\text{total}}} > \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

3.2 Ablation Tests and Baseline Comparison

We conduct ablation studies to examine the contributions of feature selection and uncertainty modelling. Specifically, we evaluate models with MC dropout only, Gaussian likelihood only, and the combination of both uncertainty components.

To assess whether the models capture meaningful predictive signals, we compare them against deterministic and benchmark strategies. We first consider a deterministic LSTM model as a baseline to evaluate the impact of incorporating predictive uncertainty. In addition, a GBP/USD buy-and-hold strategy is included as a market benchmark representing passive exposure, while a random trading strategy serves as a naive lower bound. We further compare deterministic and probabilistic models under the same evaluation framework. Probabilistic models apply an SNR-based decision rule, whereas deterministic models rely on raw predictions. Performance is evaluated using both statistical and financial metrics.

4 Results

4.1 Predictive Accuracy and Uncertainty Calibration

We first evaluate whether the proposed models capture predictive signals beyond naive benchmarks. Table 1 reports predictive accuracy in terms of RMSE and MAE. All deep learning models show similar performance, with DC-LSTM, LSTM, and Bi-LSTM achieving nearly identical error levels, suggesting limited gains from architectural variations. In contrast, the Random Walk baseline performs substantially worse, indicating the presence of exploitable structure in the data. These results suggest that point-wise error metrics alone may not sufficiently distinguish model performance.

We evaluate probabilistic calibration using PICP (Prediction Interval Coverage Probability) and MPIW (Mean Prediction Interval Width) [8], which measure interval coverage and sharpness, respectively. The DC-LSTM model provides the most balanced uncertainty profile, achieving near-nominal coverage with relatively narrow prediction intervals, indicating reliable and sharp estimates. In contrast, the Dual-Branch and Bi-LSTM models achieve higher coverage at the cost of wider intervals, suggesting more conservative uncertainty estimates and less selective signals. We select the probabilistic DC-LSTM (Gaussian + MC Dropout) as our final model, as it offers the best balance between calibration and sharpness and is most suitable for downstream trading evaluation.

4.2 Strategy Performance and Backtesting

Trading performance is evaluated under the SNR-based strategy using standard investment metrics, including total return, Sharpe ratio, hit rate, and market exposure [10]. The results are shown in Table 2.

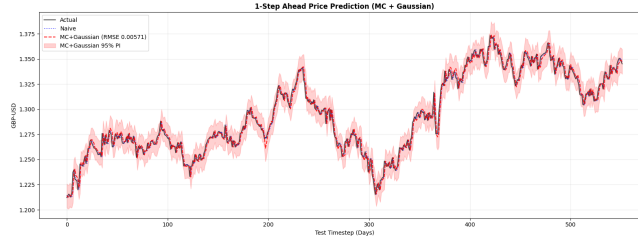


Fig. 2: One-step-ahead price prediction with Gaussian uncertainty bands. The figure compares the realised GBP/USD price path with the predictive mean and the associated 95% interval.

Table 1: Predictive Accuracy and Uncertainty Metrics

Model Architecture	RMSE (Price)	MAE (Price)	PICP (Coverage)	MPIW (Width)
DC-LSTM (Probabilistic)	0.0057	0.0043	0.9529	0.0240
DC-LSTM (Deterministic)	0.0057	0.0043	N/A	N/A
LSTM (Probabilistic)	0.0058	0.0044	0.9601	0.0248
LSTM (Deterministic)	0.0057	0.0042	N/A	N/A
Zero Baseline	0.0057	0.0042	N/A	N/A
Random Walk	0.0080	0.0066	N/A	N/A
Dual Branch (Probabilistic)	0.0061	0.0045	0.9837	0.0345
Bi-LSTM (Probabilistic)	0.0057	0.0043	0.9837	0.0357

Table 2: Trading Performance Comparison Under Different Strategies

Strategy	Total Return	Sharpe Ratio	Hit Rate	Market Exposure
DC-LSTM (Filtered)	25.79%	1.84	51.46%	61.96%
DC-LSTM (Deterministic)	22.76%	1.34	52.17%	100.00%
LSTM (Filtered)	16.15%	1.29	52.82%	64.13%
LSTM (Deterministic)	13.50%	0.83	50.72%	100.00%
Buy & Hold (Market)	10.76%	0.67	N/A	100.00%
Random Walk (Naive Baseline)	0.85%	0.02	49.24%	100.00%

The backtesting results show that uncertainty-aware filtering materially improves risk-adjusted performance. For DC-LSTM, the deterministic strategy achieves a total return of 22.76% with a Sharpe ratio of 1.34 under full market exposure. Once uncertainty filtering is applied, the total return increases to 25.79%, while the Sharpe ratio improves substantially to 1.84 and market exposure drops sharply to 61.96%. This indicates that the filtered strategy is able to generate higher returns while participating in the market less frequently, thereby delivering more efficient risk-adjusted performance.

Compared with LSTM, DC-LSTM consistently achieves stronger performance, suggesting that the DC-LSTM architecture is more effective at extracting tradable signals when combined with uncertainty filtering. Importantly, hit rates across all strategies remain close to 50–53%, with DC-LSTM (Filtered) achieving 51.46% compared to 52.17% for its deterministic version. This indicates that improvements in trading performance are not driven by directional accuracy, but rather by better trade selection enabled by uncertainty estimates. In addition, the proposed models substantially outperform benchmark strategies such as buy-and-hold and random walk. These findings are further supported by Fig. 3, where the uncertainty-aware strategy exhibits smoother growth and sustained performance.

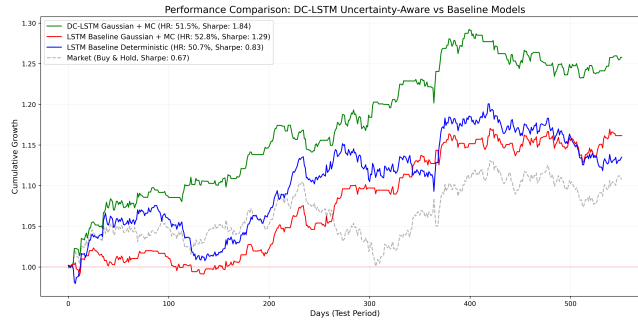


Fig. 3: Performance comparison of uncertainty-aware and deterministic trading strategies (DC-LSTM, LSTM) against the market benchmark.

5 Discussion and Conclusion

The results suggest that the main value of the proposed framework lies not in substantially improving point forecasting accuracy, but in turning weak signals into more useful trading decisions. In this FX setting, where daily returns are highly noisy, RMSE and MAE remain close across models and baselines as expected. However, once uncertainty is incorporated into the decision rule, the model achieves better risk-adjusted performance by reducing unnecessary market exposure.

Another important finding is that useful uncertainty is not simply a matter of producing wider prediction intervals. What matters is whether the uncertainty estimate is well calibrated and still informative for decision-making. In trading, overly conservative intervals may reduce actionable opportunities, while overconfident predictions may lead to unnecessary risk.

There are also several limitations to the current study. The experiments focus on a single FX pair and a limited out-of-sample period, so generalisability across other assets and market regimes remains unclear. Additionally, the backtesting setting does not fully account for practical frictions such as transaction costs or slippage, and the uncertainty estimates from Gaussian likelihood and MC Dropout are only approximations, and may be less reliable under strong non-stationarity. Future work could therefore test the approach on a broader set of assets, incorporate more realistic trading assumptions, and compare it with other uncertainty-aware forecasting methods.

In conclusion, FX forecasting remains challenging because return series are highly noisy and weakly predictable. Our results show that uncertainty-aware DC-LSTM does not substantially improve point prediction accuracy, but it does improve trading performance by filtering low-confidence signals.

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